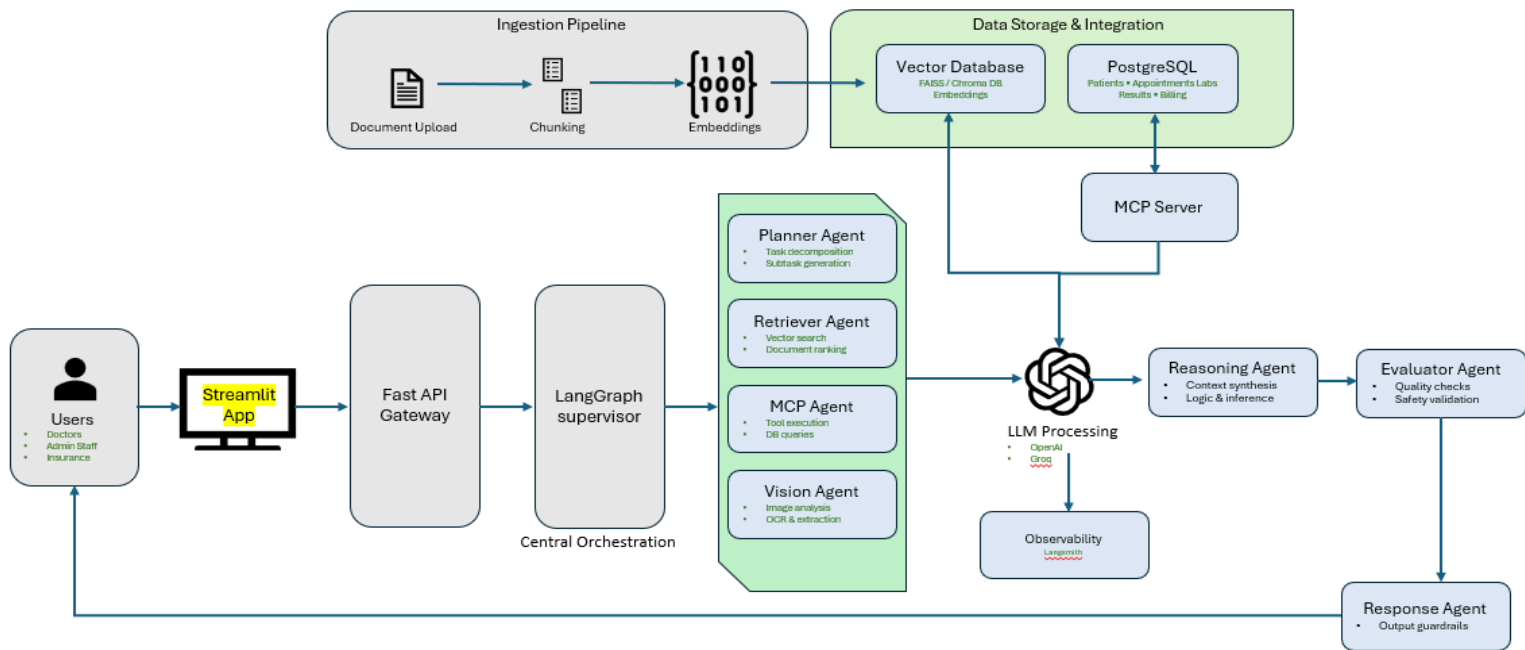


ARCHITECTURE BREAKDOWN



Entry Point — Users & Frontend

- Users (Doctors, Admin Staff, Insurance) interact via a Streamlit App
- Requests go through a FastAPI Gateway as the backend entry point

Frontend:

1. Streamlit App

This is frontend. It is like the hospital reception desk.

Users can: Ask questions, Upload PDFs, Upload prescriptions, Upload reports.

It doesn't do any heavy thinking. It simply collects information.

2. FastAPI

The FastAPI backend receives requests from the Streamlit app. Its job is to: Accept the user's request, Pass it to the AI system & Return the final response. It acts like a bridge between the frontend and the AI backend.

3. Central Orchestration — LangGraph Supervisor

The brain of the system. It coordinates 4 specialized agents:

Agent	Role
Planner Agent	Breaks user queries into tasks & subtasks
Retriever Agent	Does vector search & ranks relevant documents
MCP Agent	Executes tools & runs DB queries
Vision Agent	Handles image analysis & OCR extraction

4. Ingestion Pipeline

- Documents are uploaded → chunked → converted into embeddings
- Stored in a Vector Database (FAISS/Chroma) for semantic search

5. Data Storage & Integration

- Vector DB — stores embeddings for RAG-based retrieval
- PostgreSQL — stores structured data (Patients, Appointments, Labs, Results, Billing)
- MCP Server — bridges agents to both databases

6. LLM Processing & Reasoning

- Powered by OpenAI and Groq models
- Feeds into a Reasoning Agent for context synthesis & logic/inference
- Evaluator Agent — quality checks & safety validation
- Response Agent — applies output guardrails before returning to the user

7. Observability

This component monitors the entire workflow.

It tracks:

- Which agents were used.
- Time taken.
- Errors.
- System performance.

It helps developers debug and improve the system.

Observability via LangSmith for monitoring & tracing

8. Workflow

The user interacts with the Streamlit application by asking a question or uploading a document or medical image. The request is sent to the FastAPI backend, where the LangGraph supervisor analyzes it and assigns tasks to the appropriate agents. The Planner Agent breaks down the request, the Retriever Agent searches hospital documents stored in the vector database, the MCP Agent fetches structured patient data from PostgreSQL through the MCP server, and the Vision Agent processes uploaded images if required. The LLM combines all the retrieved information, while the Reasoning Agent generates a contextual response. Finally, the Evaluation Agent checks the response for accuracy and safety before the Response Agent delivers the final output to the user.